

D369

Finance Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A company's revenue grew from \$40.0 million in 2020 to \$58.564 million in 2024. The firm moved its headquarters in 2022 and hired a new CFO. What is the compound annual growth rate?

- A. 11.6%
- B. 46.4%
- C. 10.0%
- D. 7.9%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Revenue of \$50 million is expected to grow at a CAGR of 8% for 3 years. What revenue is expected at the end of year 3?

- A. About \$39.69 million
- B. \$54.0 million
- C. About \$62.99 million
- D. \$62.0 million

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An analyst models a project that costs \$100,000 today and returns \$40,000 at the end of each of the next 3 years, at an 8% rate. She enters =NPV(8%, -100000, 40000, 40000, 40000). The model is color-coded and saved on a shared drive. What is wrong?

- A. It discounts the \$100,000 outlay by a year; use =NPV(8%, the three inflows) - 100000, which gives about \$3,084.
- B. NPV() cannot handle outlays; only IRR() works for this.
- C. Nothing; the result of about \$2,855 is correct.
- D. The color coding and shared-drive links distort the result.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An analyst wants to value a project whose first cash flow happens today, using Excel's NPV() function. Which approach is correct?

- A.** Put every cash flow, including time 0, inside NPV().
- B.** Rely on NPV() treating the first value as happening today.
- C.** Never use NPV(); only XNPV() works.
- D.** Leave the time-0 cash flow out of NPV() and add it separately.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A firm has EBIT of \$20 million, a 25% tax rate, depreciation of \$4 million, capital expenditures of \$6 million and an increase in net working capital of \$2 million. It paid \$3 million in interest and has 900 employees. What is free cash flow to the firm?

- A.** \$13 million
- B.** \$11 million
- C.** \$3 million
- D.** \$8 million

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A firm's target weights are 30% debt and 70% equity. Its pre-tax cost of debt is 6%, its tax rate is 21% and its cost of equity is 10%. The company is headquartered in Texas. What is its WACC?

- A.** About 8.42%
- B.** About 6.32%
- C.** 8.00%
- D.** 8.80%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst changes only the sales growth input from 5% to 3%, holding all other inputs fixed, and sees NPV fall from \$4 million to \$1 million. The model has 14 tabs and uses blue font for inputs. Which technique is this?

- A. Sensitivity analysis
- B. Proof that the project should be rejected
- C. Monte Carlo simulation
- D. Scenario analysis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a 5-year DCF, the analyst projects year-5 free cash flow of \$10 million, long-term growth of 3% and a WACC of 8%. The firm's cost of equity is 11%. What is the terminal value at the end of year 5?

- A. \$128.75 million
- B. \$125 million
- C. \$206 million
- D. \$200 million

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A terminal value of \$150 million was computed as next-year FCF ÷ (WACC – g), with WACC of 9% and next-year FCF of \$6 million. What growth rate was assumed?

- A. 13%
- B. 5%
- C. 4%
- D. 9%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A DCF gives an enterprise value of \$500 million. The firm has \$120 million of debt, \$30 million of cash and 20 million shares. It operates 11 factories. What is the value per share?

- A. \$17.50
- B. \$19.00
- C. \$25.00
- D. \$20.50

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

A firm's equity is worth \$300 million. It has \$100 million of debt and \$40 million of cash. What is its enterprise value?

- A. \$400 million
- B. \$360 million
- C. \$440 million
- D. \$240 million

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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12.

In cell C5 an analyst types =B5*\$B\$1, where B1 holds the tax rate. The sheet has a dark-blue header row and 36 monthly columns. She copies the formula down to C6. What does C6 contain?

- A. =B5*\$B\$1
- B. =B6*B2
- C. =C6*\$C\$1
- D. =B6*\$B\$1

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A company's revenue grew from \$40.0 million in 2020 to \$58.564 million in 2024. ~~The firm moved its headquarters in 2022 and hired a new CFO.~~ What is the compound annual growth rate?

Target word: CAGR **Decides:** Four growth periods; ratio 1.4641.

A	Adjacent category	Total growth $46.4\% \div 4$, a simple average that ignores compounding.
B	Dropped element	Total growth; drops annualizing.
C	Correct	$1.4641^{(1/4)} - 1 = 10\%$.
D	Wrong stem number	Uses 5 periods by counting the years listed.

2. Answer C Concept 1 • transfer

Revenue of \$50 million is expected to grow at a CAGR of 8% for 3 years. What revenue is expected at the end of year 3?

Target word: solve for ending value **Decides:** $\$50M \times 1.08^3$.

A	Inverted operation	Discounts instead of compounding.
B	Wrong stem number	One year of growth.
C	Correct	$\$50M \times 1.2597$.
D	Dropped element	Simple growth (24%) drops compounding.

3. Answer A Concept 2

An analyst models a project that costs \$100,000 today and returns \$40,000 at the end of each of the next 3 years, at an 8% rate. She enters =NPV(8%, -100000, 40000, 40000, 40000). ~~The model is color coded and saved on a shared drive.~~ What is wrong?

Target word: formula error **Decides:** NPV() discounts the first value a full period.

A	Correct	Time-0 cash belongs outside NPV().
B	Absolute language	NPV works when the outlay is added separately.
C	Dropped element	Ignores Excel's end-of-period timing.
D	Camouflage	Formatting and storage do not change the math.

4. Answer D Concept 2 • transfer

An analyst wants to value a project whose first cash flow happens today, using Excel's NPV() function. Which approach is correct?

Target word: correct setup **Decides:** NPV() assumes the first value is one period out.

A	Inverted operation	Discounts today's cash by a period.
B	Confusion set	NPV() assumes end-of-period timing.
C	Absolute language	NPV() works with correct setup.
D	Correct	Keeps today's cash undiscounted.

5. Answer B Concept 3

A firm has EBIT of \$20 million, a 25% tax rate, depreciation of \$4 million, capital expenditures of \$6 million and an increase in net working capital of \$2 million. ~~It paid \$3 million in interest and has 900 employees.~~ What is free cash flow to the firm?

Target word: FCFF **Decides:** $\text{EBIT}(1 - T) = \$15\text{M}$.

A	Dropped element	Leaves out the working-capital increase.
B	Correct	$\$15\text{M} + \$4\text{M} - \$6\text{M} - \2M .
C	Inverted operation	Subtracts depreciation instead of adding it back.
D	Camouflage	Subtracts interest, which FCFF excludes.

6. Answer A Concept 4

A firm's target weights are 30% debt and 70% equity. Its pre-tax cost of debt is 6%, its tax rate is 21% and its cost of equity is 10%. ~~The company is headquartered in Texas.~~ What is its WACC?

Target word: WACC **Decides:** After-tax debt = $6\% \times 0.79 = 4.74\%$.

A	Correct	$0.3 \times 4.74\% + 0.7 \times 10\% = 1.42\% + 7.0\%$.
B	Inverted operation	Swaps the weights.
C	Adjacent category	A simple average of 6% and 10%.
D	Dropped element	Leaves out the tax shield.

7. Answer A Concept 5

An analyst changes only the sales growth input from 5% to 3%, holding all other inputs fixed, and sees NPV fall from \$4 million to \$1 million. ~~The model has 14 tabs and uses blue font for inputs.~~ Which technique is this?

Target word: technique **Decides:** One input changed, others held constant.

A	Correct	Tests how one variable moves the result.
B	Absolute language	NPV is still positive; sensitivity shows risk, not a verdict.
C	Adjacent category	Draws many random combinations of inputs.
D	Confusion set	Scenarios change several inputs together.

8. Answer C Concept 6

In a 5-year DCF, the analyst projects year-5 free cash flow of \$10 million, long-term growth of 3% and a WACC of 8%. ~~The firm's cost of equity is 11%.~~ What is the terminal value at the end of year 5?

Target word: terminal value **Decides:** $\text{FCF}_6 = \$10.3\text{M}$; divide by $8\% - 3\%$.

A	Camouflage	Uses the cost of equity ($11\% - 3\%$).
B	Adjacent category	Zero-growth perpetuity ($\$10\text{M} \div 0.08$).
C	Correct	$\$10.3\text{M} \div 0.05$.
D	Dropped element	Skips growing year-5 FCF one more year.

9. Answer B Concept 6 • transfer

A terminal value of \$150 million was computed as next-year FCF ÷ (WACC – g), with WACC of 9% and next-year FCF of \$6 million. What growth rate was assumed?

Target word: solve for g **Decides:** $\text{WACC} - g = \$6\text{M} \div \$150\text{M} = 4\%$.

A	Inverted operation	Adds the spread to WACC.
B	Correct	$9\% - 4\%$.
C	Dropped element	The spread, not g.
D	Wrong stem number	The WACC.

10. Answer D Concept 7

A DCF gives an enterprise value of \$500 million. The firm has \$120 million of debt, \$30 million of cash and 20 million shares. ~~It operates 11 factories.~~ What is the value per share?

Target word: per-share equity value **Decides:** $\text{Equity} = \$500\text{M} - \$120\text{M} + \$30\text{M} = \410M .

A	Inverted operation	Subtracts cash instead of adding it.
B	Dropped element	Leaves out the cash.
C	Adjacent category	Treats EV as equity value.
D	Correct	$\$410\text{M} \div 20\text{M}$.

11. Answer B Concept 7 • transfer

A firm's equity is worth \$300 million. It has \$100 million of debt and \$40 million of cash. What is its enterprise value?

Target word: solve for EV **Decides:** $\text{EV} = \text{equity} + \text{debt} - \text{cash}$.

A	Dropped element	Leaves out cash.
B	Correct	$\$300\text{M} + \$100\text{M} - \$40\text{M}$.
C	Inverted operation	Adds cash instead of subtracting.
D	Confusion set	Runs the equity-value bridge in the wrong direction.

12. Answer D Concept 8

In cell C5 an analyst types $=B5*\$B\1 , where B1 holds the tax rate. ~~The sheet has a dark blue header row and 36 monthly columns.~~ She copies the formula down to C6. What does C6 contain?

Target word: copied formula **Decides:** B5 is relative; $\$B\1 is locked.

A	Confusion set	Treats every reference as absolute.
B	Dropped element	Ignores the \$ locks.
C	Adjacent category	Shifts columns, which only happens when copying across.
D	Correct	The relative row shifts; the absolute reference stays.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Compound annual growth rate	$CAGR = (\text{ending} \div \text{beginning})^{(1 \div \text{years})} - 1$; years = number of growth periods, not data points.	Averaging total growth by years, or counting data points as periods.	A start value, an end value and a range of years.
2	Spreadsheet NPV timing	Excel's NPV() discounts every value, starting one period out; add the time-0 cash flow outside the function.	Putting the upfront cost inside NPV().	A formula that includes the initial outlay in NPV().
3	Free cash flow to the firm	$FCFF = EBIT \times (1 - \text{tax rate}) + \text{depreciation} - \text{capital expenditures} - \text{increase in net working capital}$.	Subtracting interest, or dropping working capital.	EBIT, tax, depreciation, capex, NWC change and interest listed together.
4	Weighted average cost of capital	$WACC = W_e \times R_e + W_d \times R_d \times (1 - T)$ using target or market weights.	Leaving out the tax shield or swapping weights.	Weights, cost of debt, cost of equity and tax rate.
5	Sensitivity vs. scenario analysis	Sensitivity changes one input while holding others constant; scenario analysis changes several inputs together.	Calling a one-variable test a scenario.	How many inputs change at once.
6	Terminal value (Gordon growth)	$TV \text{ at year } n = FCF(n) \times (1 + g) \div (WACC - g)$.	Skipping the growth step or using the cost of equity.	Final-year FCF, long-term growth and WACC.
7	Enterprise value to equity value	Equity value = enterprise value – debt + cash; divide by shares for per-share value.	Treating EV as equity value, or subtracting cash.	EV, debt, cash and share count.
8	Relative vs. absolute cell references	A \$ locks a column or row; unlocked references shift when a formula is copied.	Assuming everything shifts, or nothing does.	A formula with \$ signs copied to another cell.

Course scope (public description): This course is designed as a synthesis of the knowledge learners have acquired throughout the program. The course culminates in a performance assessment that requires learners to apply the competencies gained throughout the finance program. In this course, learners will draw upon the concepts and techniques introduced in the undergraduate finance program to perform a comprehensive financial analysis of an enterprise. In completing the course, learners will perform analyses with spreadsheet software to simulate a real-world experience of a finance career professional.